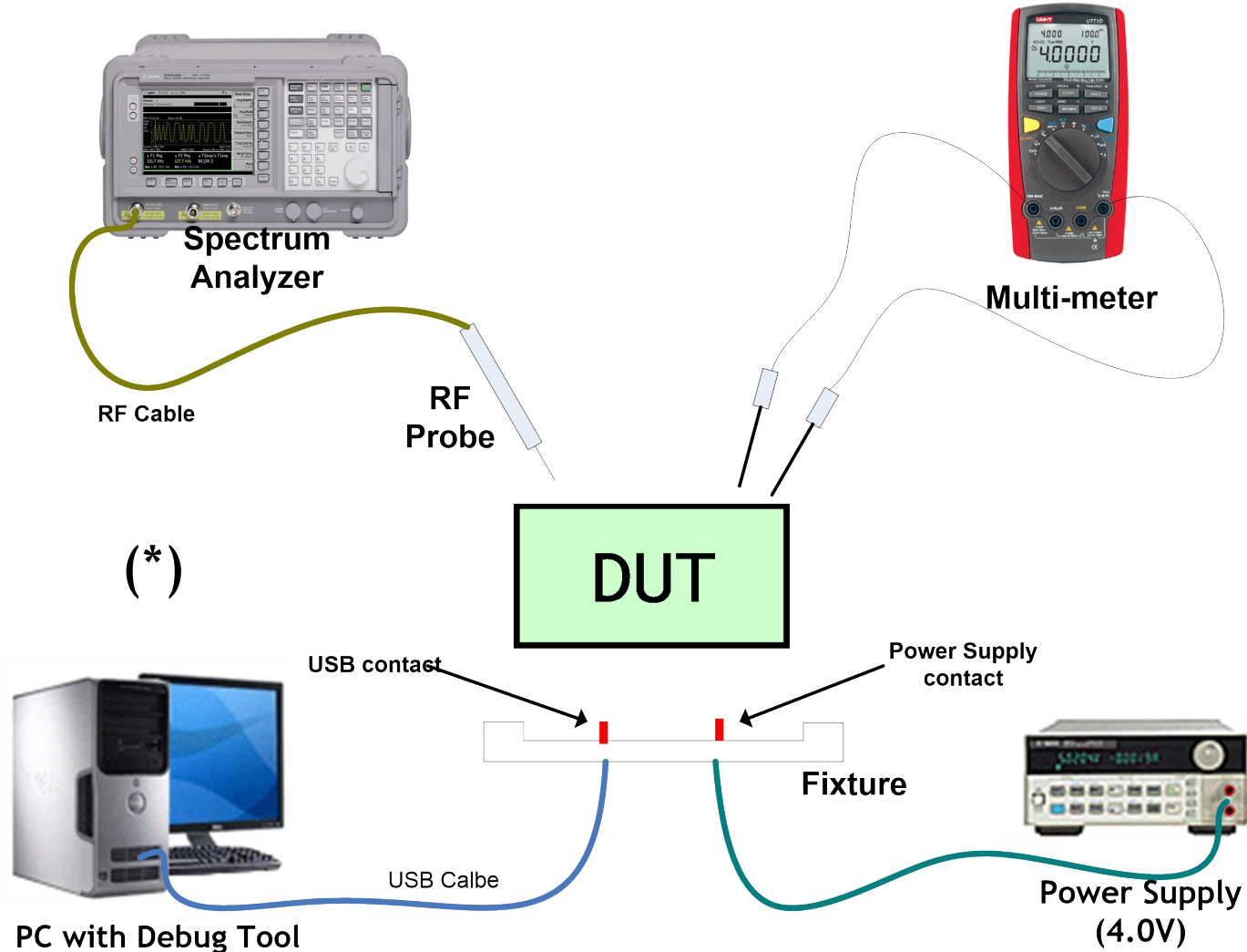


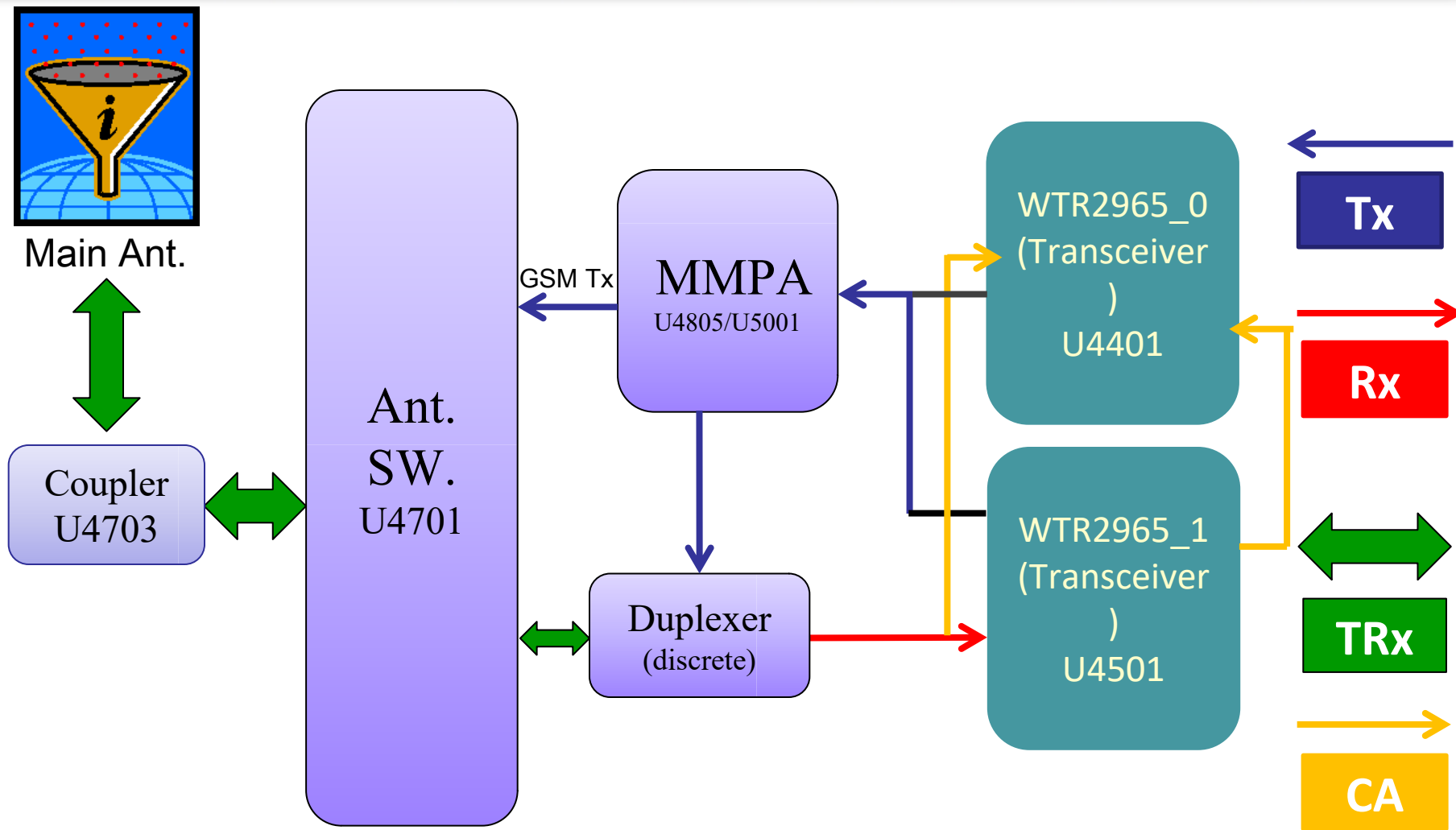
ASUS ZE520KL

RF Hardware Trouble Shooting Guide

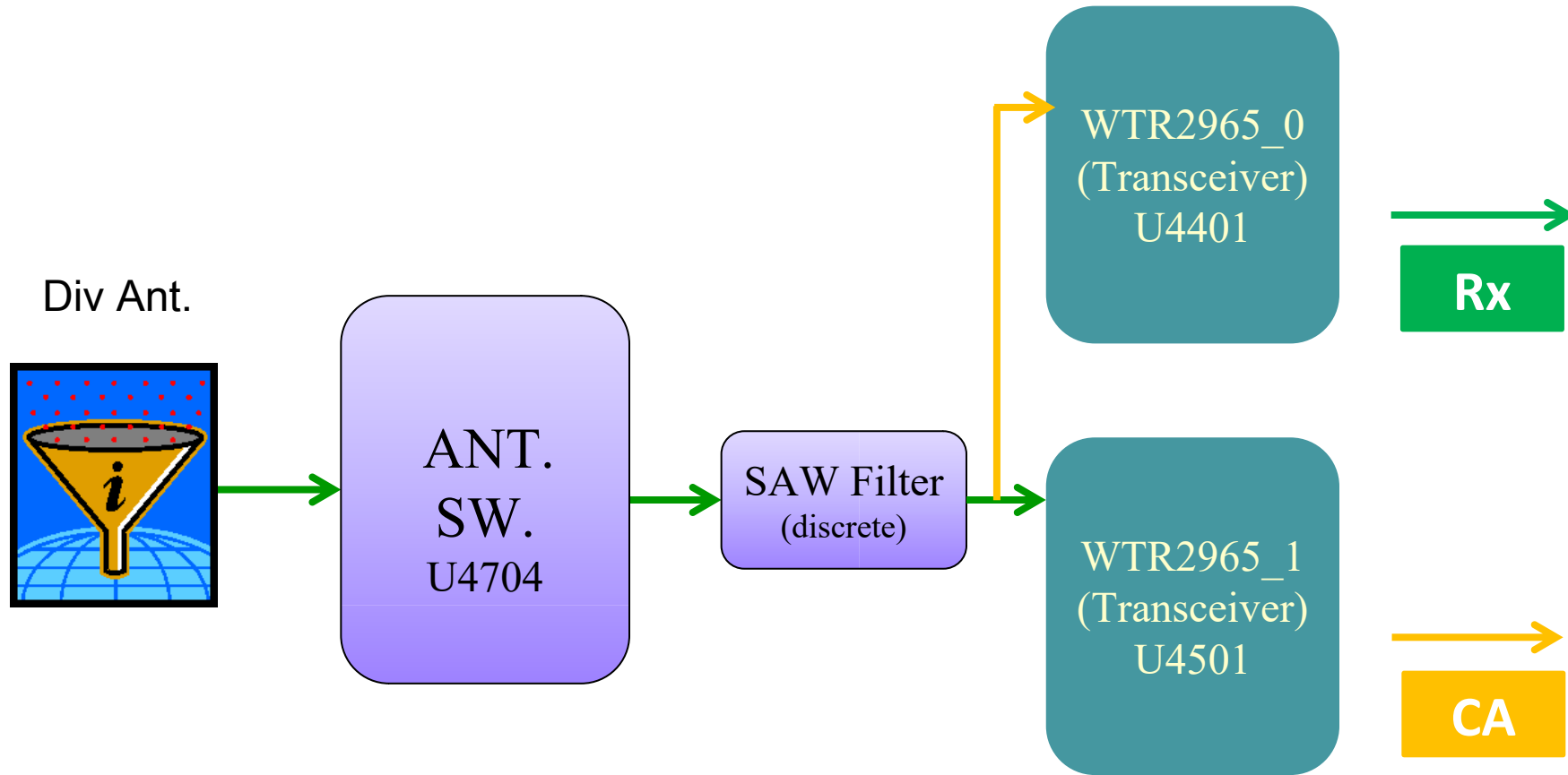
B/L Repair and Test Environment



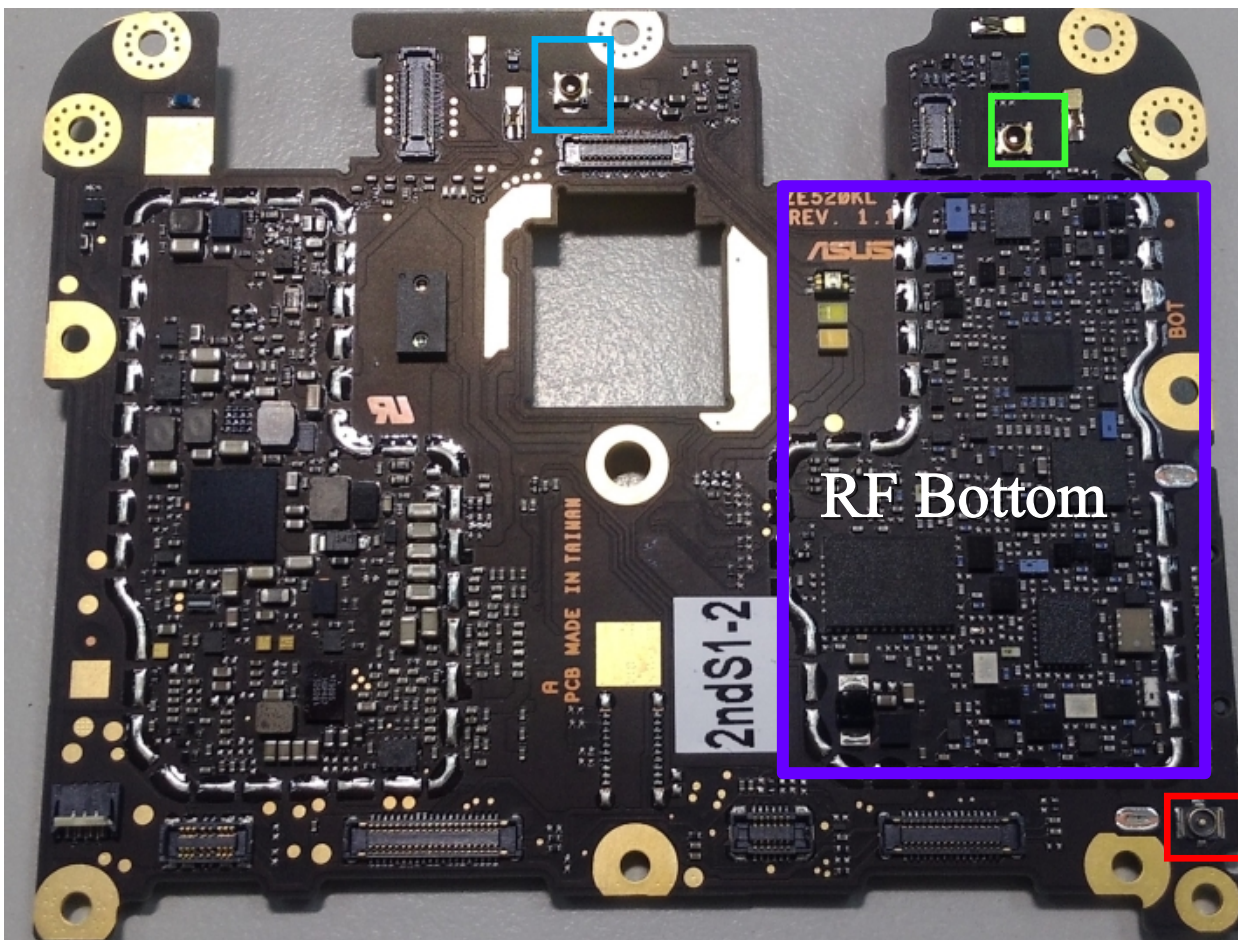
2G/3G/LTE Main Antenna Block Diagram



WCDMA/LTE Diversity Antenna Block Diagram



ZE520KL RF Portion From PCBA Bottom View



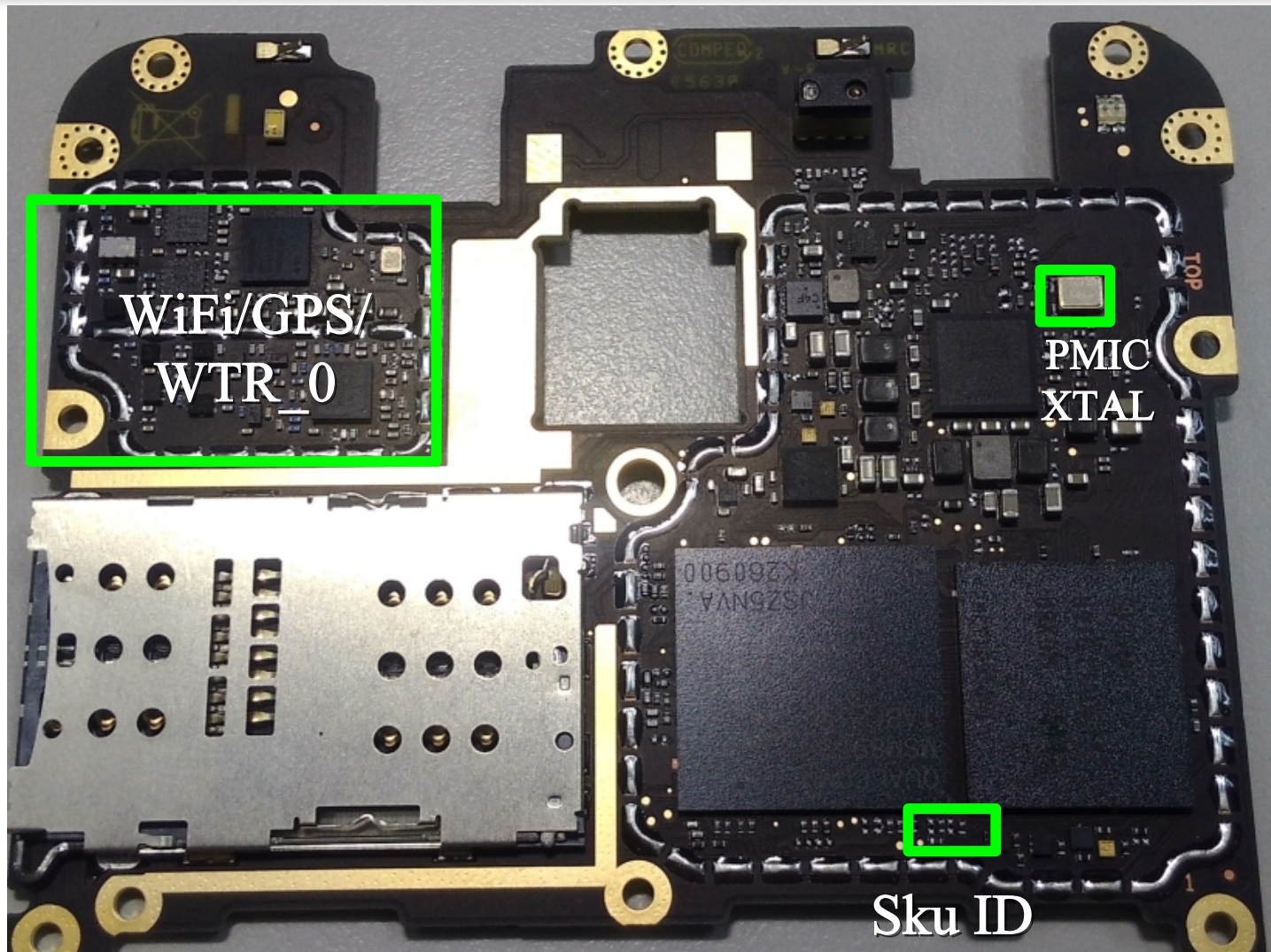
TW/JP/HK (Support CA)
GSM: 850/900/1800/1900
WCDMA: Band 1/2/5/6/8/19
FDD-LTE Band
1/2/3/5/7/8/9/18/19/26/28
TDD-LTE: 38/39/40/41

WW
GSM: 850/900/1800/1900
WCDMA: Band 1/2/5/8
FDD-LTE Band 1/2/3/5/7/8/20

ID/IN/TH
GSM: 850/900/1800/1900
WCDMA: Band 1/5/8
FDD-LTE Band 1/3/5/8
TDD-LTE: 40

-  Main TRx RF Connector
-  Diversity Rx RF Connector
-  WIFI/BT/GPS Connector

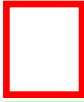
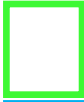
ZE520KL RF Portion From PCBA Top View

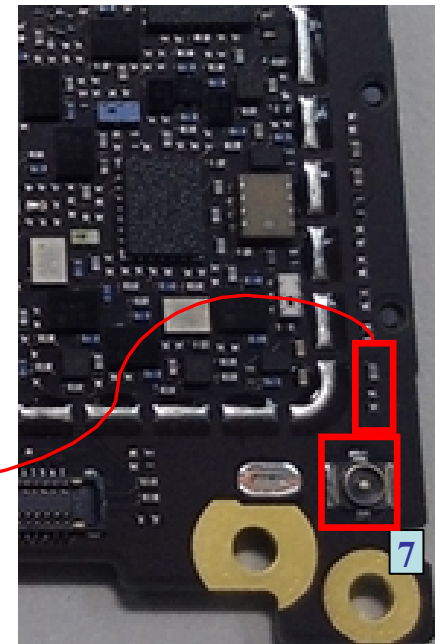
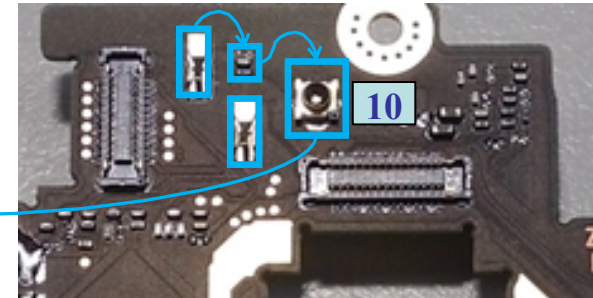
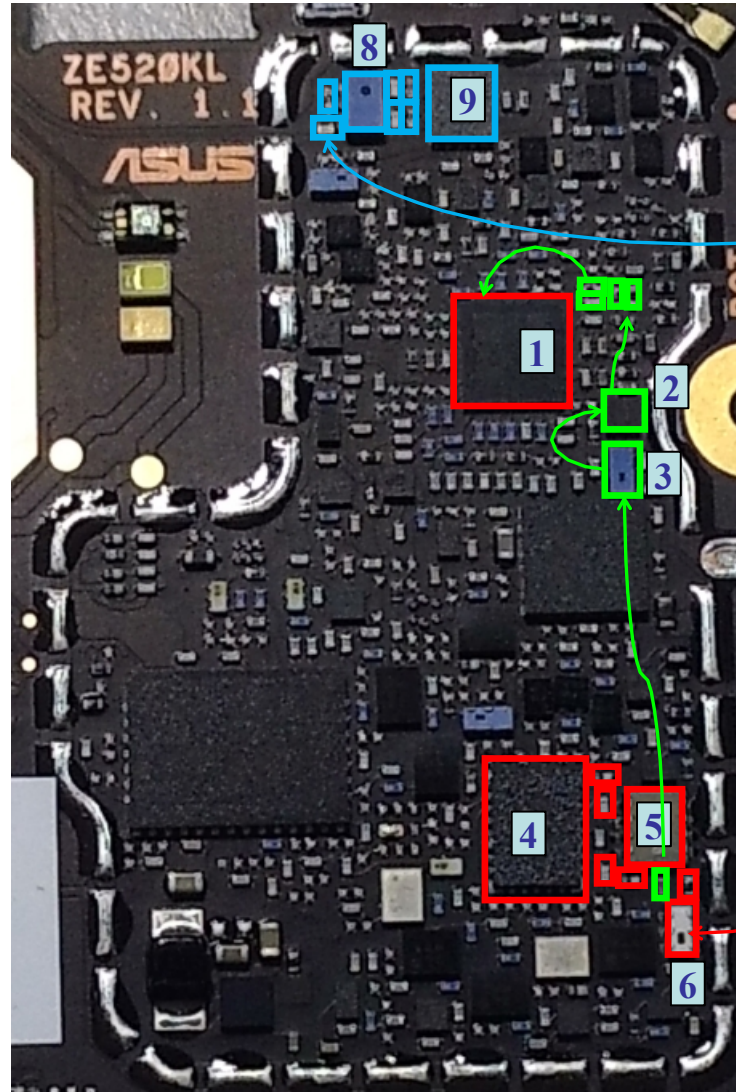


Common Comp.

RF (Bottom)

1. U4501: Transceiver(WTR2965_1)
2. U4706: SP3T for PDET
3. U4707: LPF for PDET
4. U4701: Primary Ant. SW.
5. U4702: Main Diplexer
6. U4703: Coupler
7. CON4701: Main RF Connector
8. U4705: Div Diplexer
9. U4704: Div Ant. SW.
10. CON4704: Diversity RF Connector

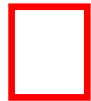
-  TRx common comp.
-  PDET common comp.
-  DRx common comp.



Common Comp.

RF (Top)

1. U4401: Transceiver(WTR2965_0)



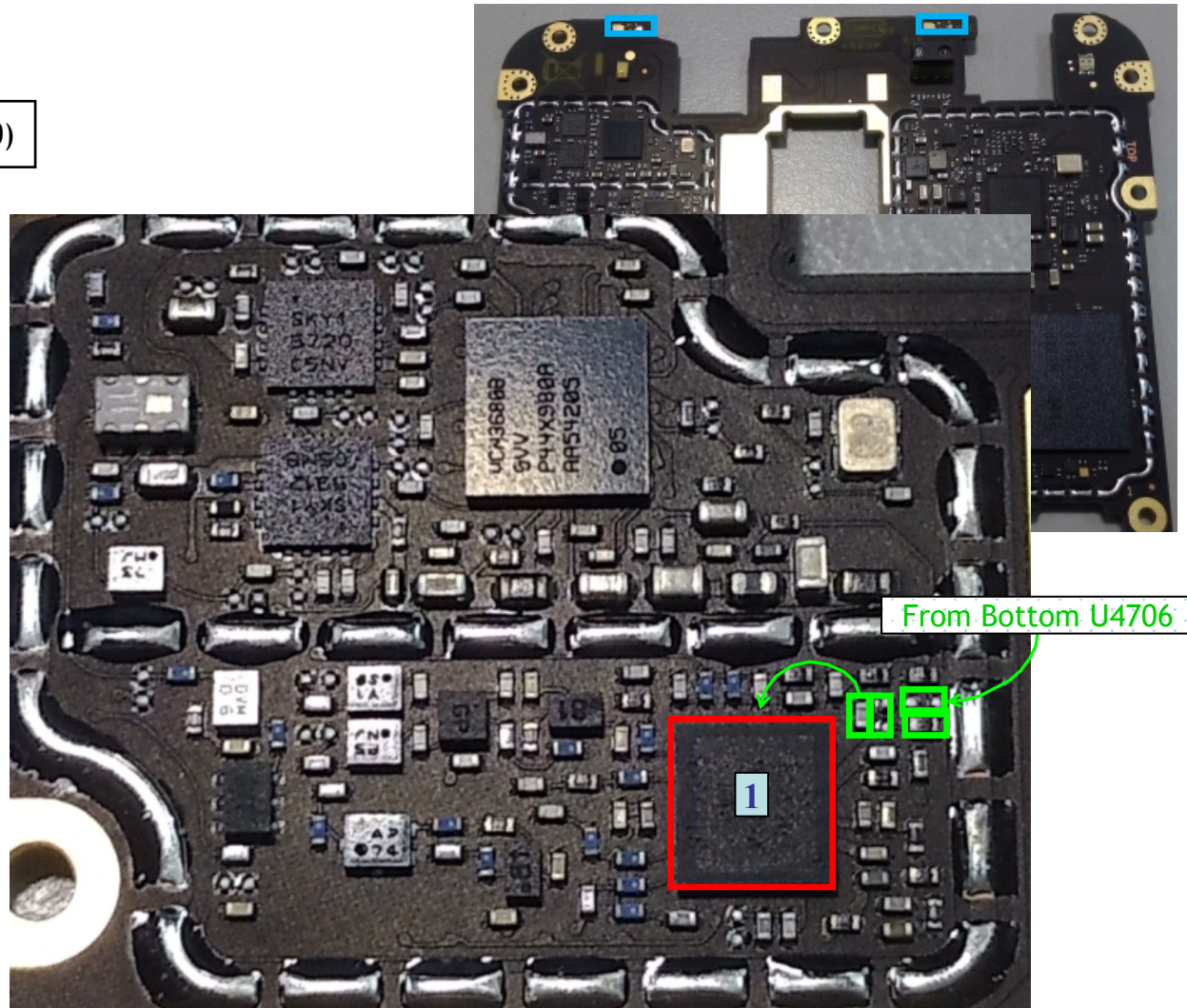
TRx common comp.



PDET common comp.



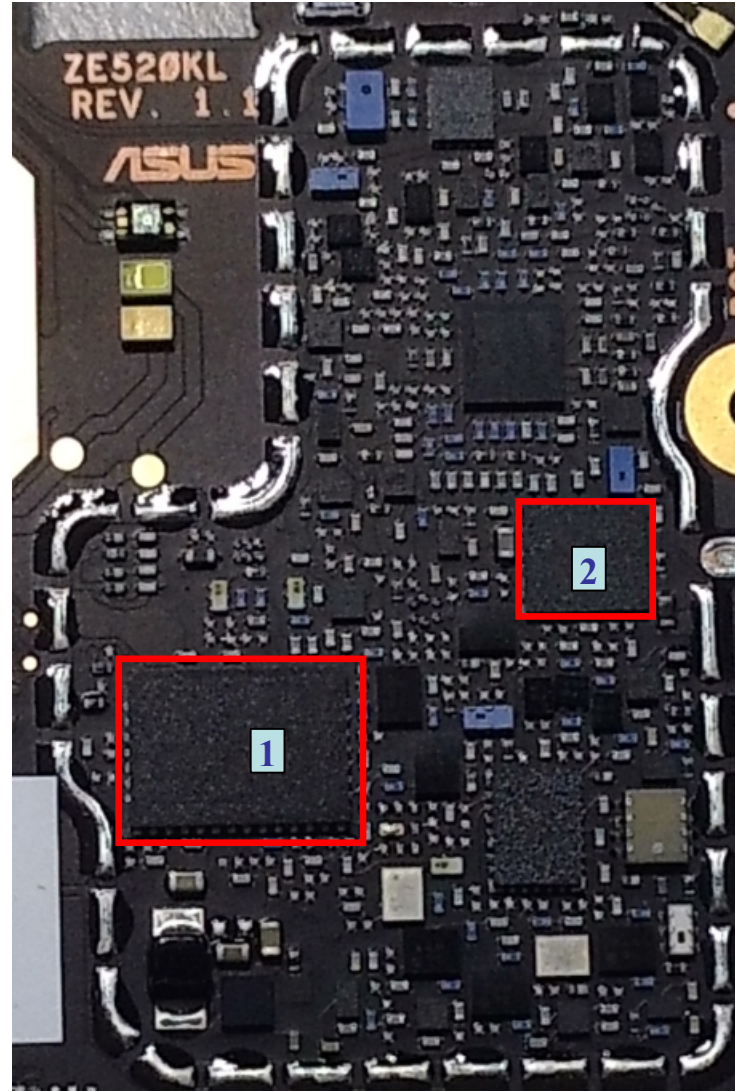
DRx common comp.



Tx Power Amplifier

RF (Bottom)

1. U4805: MMPA (LB/MB)
2. U5001: B7/B38/B39/B40/B41 PA

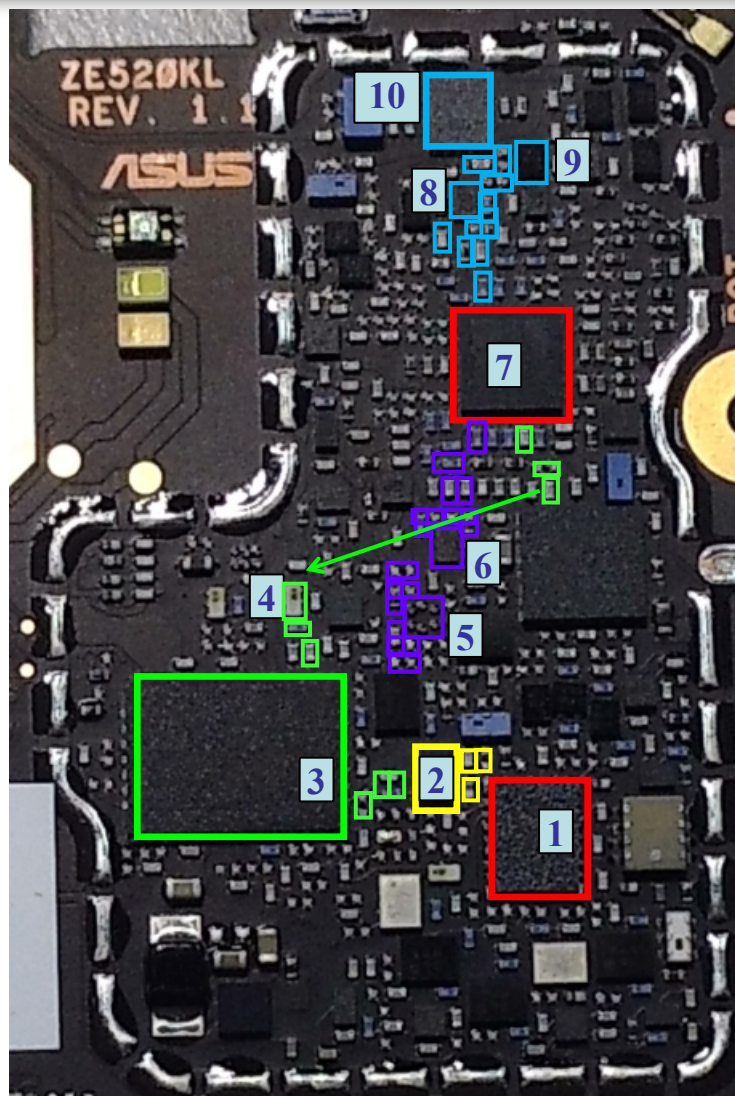


B1 TRx/DRx

RF (Bottom)

1. U4701: Primary Ant. SW.
2. U4807: B1 Duplexer
3. U4805: MMPA(MB/LB)
4. U4803: LPF for B1/B2/B3/B4/GHB
5. U4901: SPDT SW. for B1/B4
6. U4903: LNA for B1/B4
7. U4501: Transceiver(WTR2965_1)
8. U5112: LNA for B1/B4
9. U5105: B1 DRx SAW
10. U4704: Div Ant. SW.

-  TRx common comp.
-  B1 Tx Path
-  B1 TRx Path
-  B1 PRx Path
-  B1/B4 DRx Path

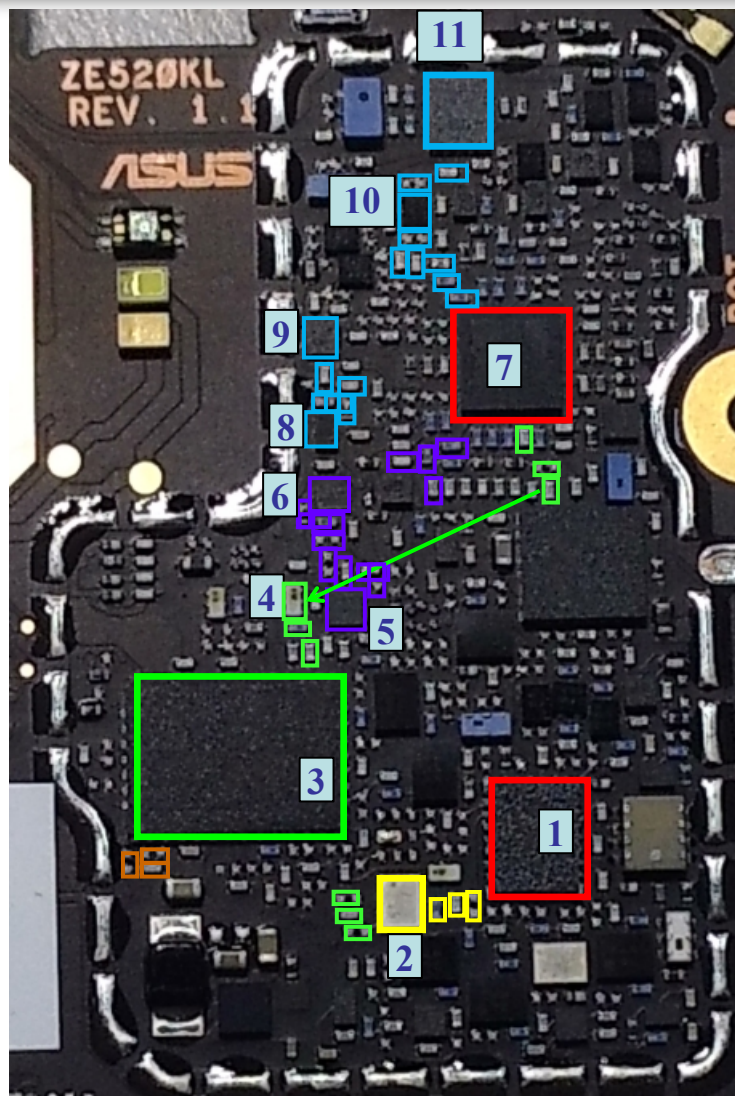


PCS/ B2 TRx/B2 DRx

RF (Bottom)

1. U4701: Primary Ant. SW.
2. U4804: B2 Duplexer
3. U4805: MMPA(MB/LB)
4. U4803: LPF for B1/B2/B3/B4/GHB
5. U4904: SP3T SW. for PRx B2/B3/B39
6. U4908: LNA for B2/B3/B39
7. U4501: Transceiver(WTR2965_1)
8. U5116: LNA for B2/B3/B39
9. U5103: SP3T SW. for DRx B2/B3/B39
10. U5123: B2 DRx SAW
11. U4704: Div Ant. SW.

-  TRx common comp.
-  B2(PCS) Tx Path
-  B2 TRx Path
-  B2 PRx (PCS Rx) Path
-  B2 DRx Path
-  PCS PAout Path

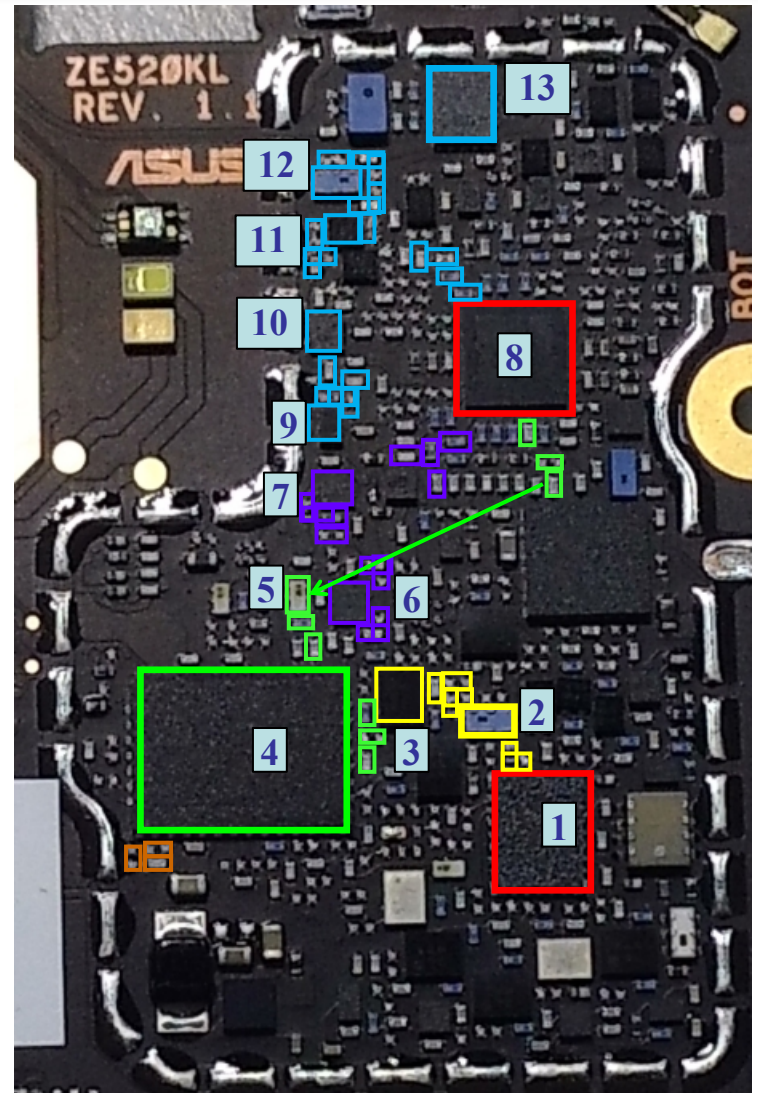


DCS/ B3 TRx/B3 DRx

RF (Bottom)

1. U4701: Primary Ant. SW.
2. U4817: B3/B7 Diplexer
3. U4802: B3 Duplexer
4. U4805: MMPA(MB/LB)
5. U4803: LPF for B1/B2/B3/B4/GHB
6. U4904: SP3T SW. for PRx B2/B3/B39
7. U4908: LNA for B2/B3/B39
8. U4501: Transceiver(WTR2965_1)
9. U5116: LNA for B2/B3/B39
10. U5103: SP3T SW. for DRx B2/B3/B39
11. U5117: B3 DRx SAW
12. U5110: B3/B7 Diplexer
13. U4704: Div Ant. SW.

- | | |
|--|--|
|  TRx common comp. |  B3 PRx (DCS Rx) Path |
|  B3(DCS) Tx Path |  B3 DRx Path |
|  B3 TRx Path |  DCS PAout Path |

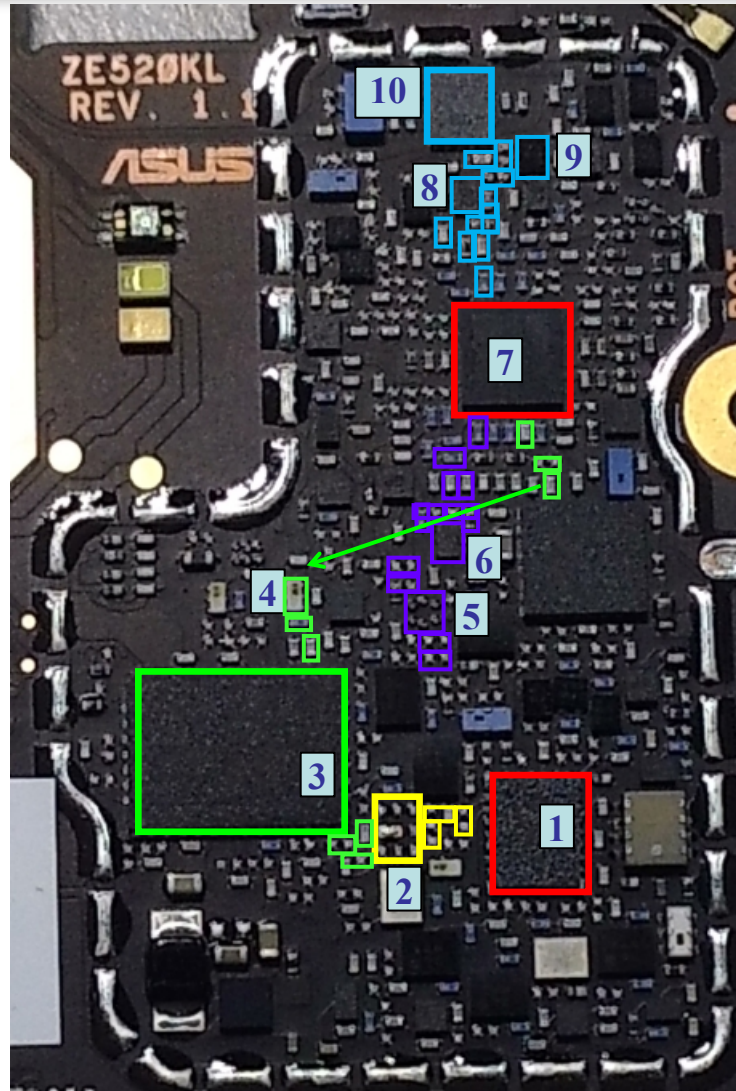


B4 TRx/DRx

RF (Bottom)

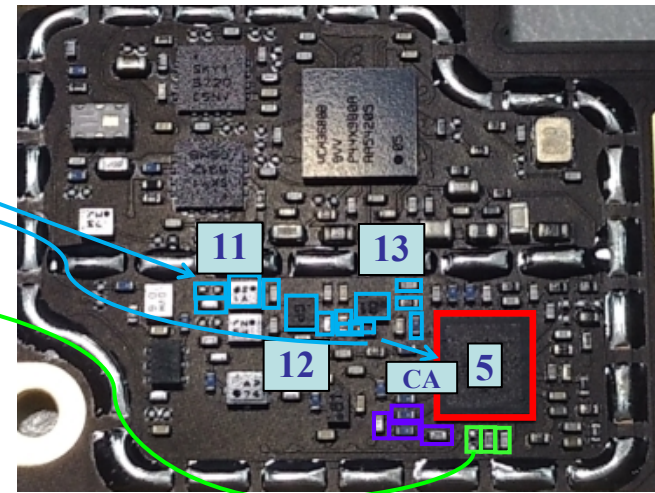
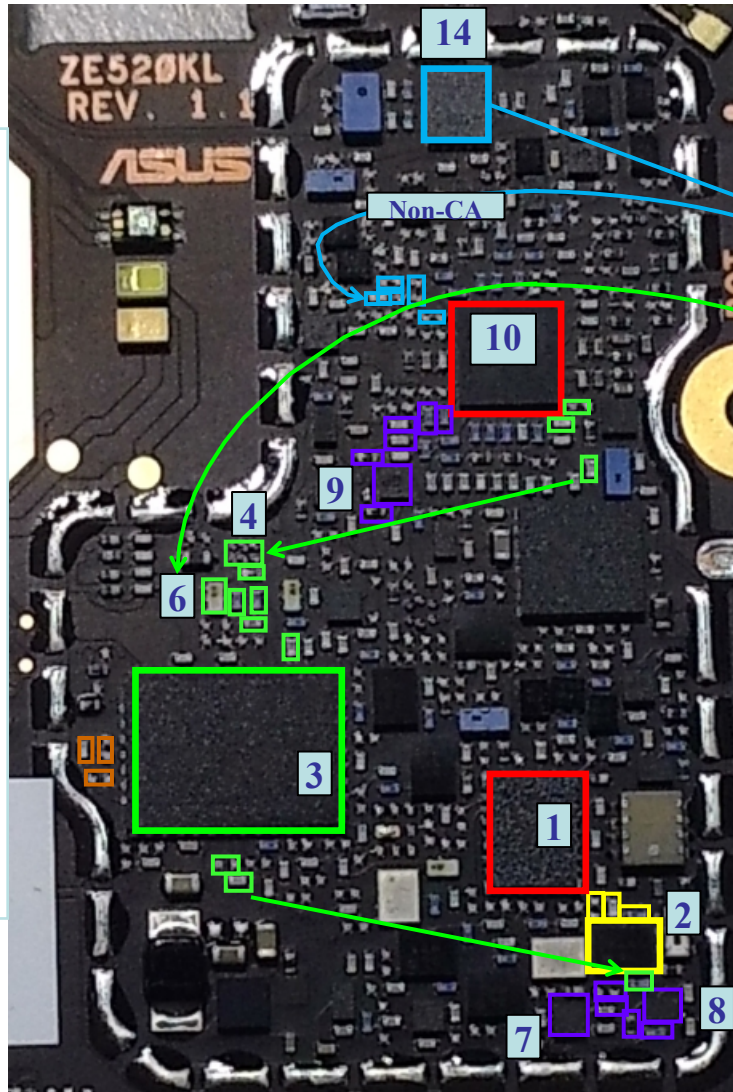
1. U4701: Primary Ant. SW.
2. U4806: B4 Duplexer
3. U4805: MMPA(MB/LB)
4. U4803: LPF for B1/B2/B3/B4/GHB
5. U4901: SPDT SW. for B1/B4
6. U4903: LNA for B1/B4
7. U4501: Transceiver(WTR2965_1)
8. U5112: LNA for B1/B4
9. U5105: B1 DRx SAW
10. U4704: Div Ant. SW.

-  TRx common comp.
-  B4 Tx Path
-  B4 TRx Path
-  B4 PRx Path
-  B1/B4 DRx Path



GSM850 / B5(B6/B18/B19/B26) TRx/ DRx

RF (Bottom)



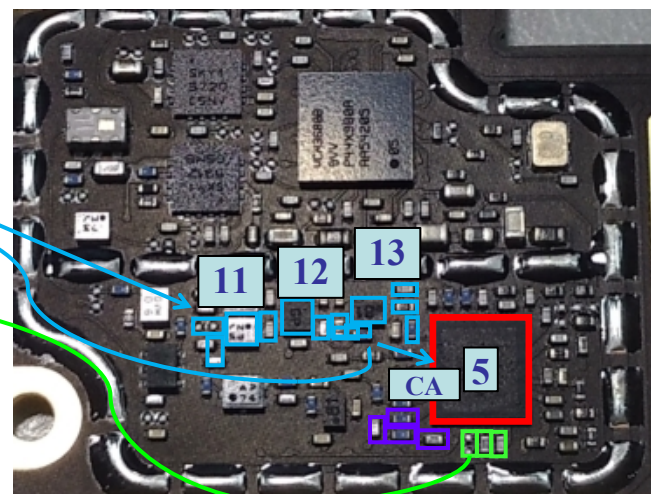
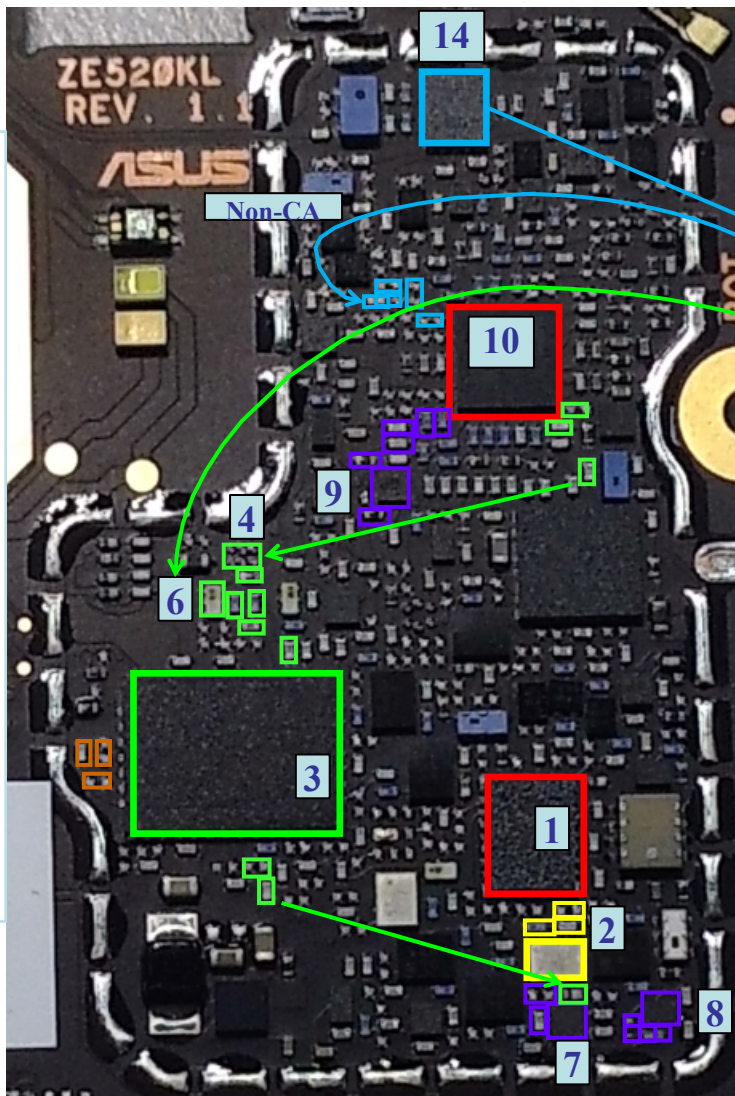
RF (Top)

- TRx common comp.
- B5(GSM850) Tx Path
- B5 TRx Path
- B5 PRx (GSM850 Rx) Path
- B5 DRx Path
- GSM850 PAout Path

GSM900 / B8 TRx/ DRx

RF (Bottom)

1. U4701: Primary Ant. SW.
2. U4809: B8 Duplexer
3. U4805: MMPA(MB/LB)
4. U4810: LPF for WTR0
B5/B8/B20/B28/GLB
5. U4401: Transceiver(WTR2965_0)
6. U4811: LPF for WTR1
B5/B8/B20/B28/GLB
7. U4911: SPDT SW. for B5/B8
8. U4913: LNA for B5/B8
9. U4906:SPDT SW. for B5/B8 to
WTR0 and WTR1
10. U4501: Transceiver(WTR2965_1)
11. U5124: B5 DRx SAW
12. U5102: SPDT SW. for B5/B8 Div
13. U5104: LNA for B5/B8
14. U4704: Div Ant. SW.

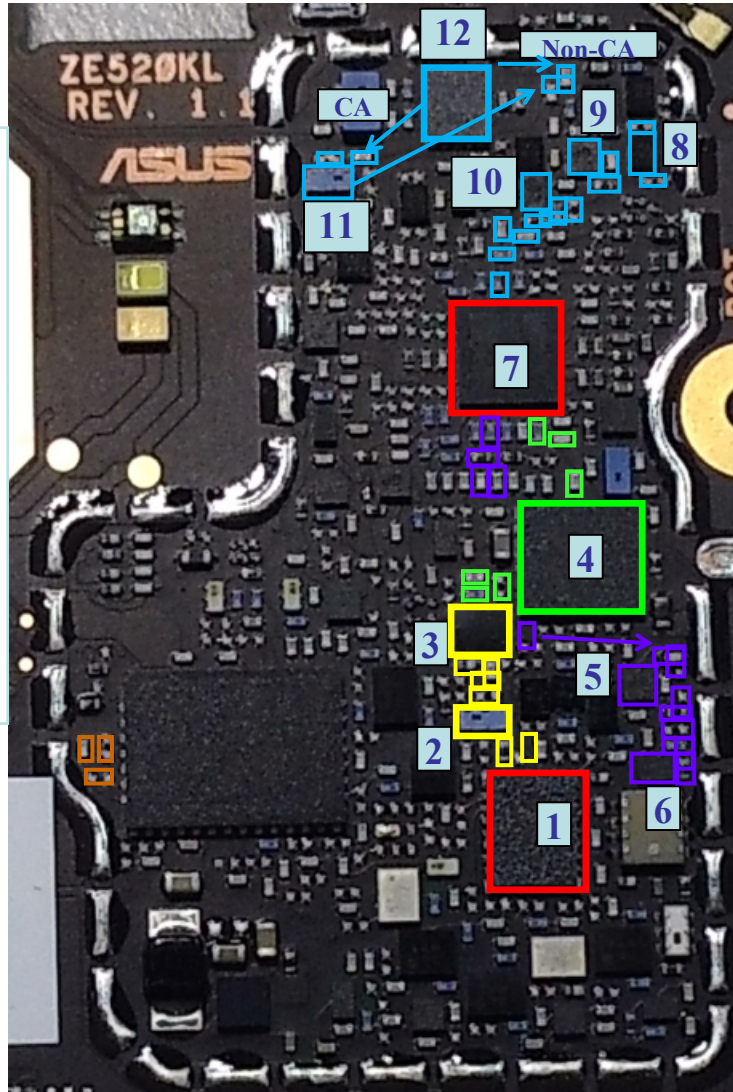


RF (Top)

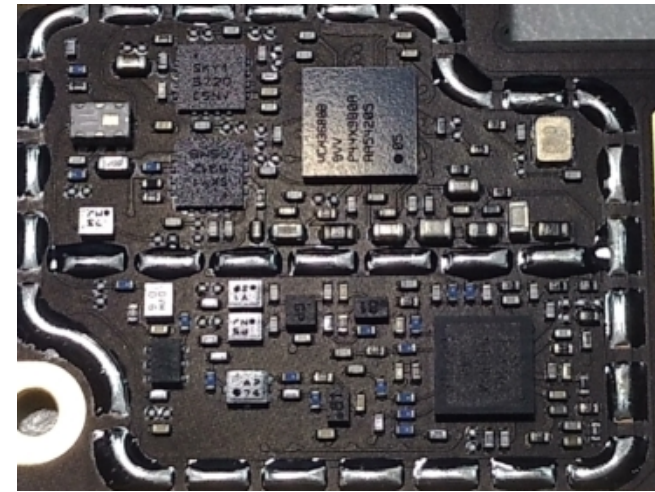
-  TRx common comp.
-  B8(GSM900) Tx Path
-  B8 TRx Path
-  B8 PRx (GSM900 Rx) Path
-  B8 DRx Path
-  GSM900 PAout Path

B7 TRx/ B7 DRx

RF (Bottom)



1. U4701: Primary Ant. SW.
2. U4817: B3/B7 Diplexer
3. U5004: B7 Duplexer
4. U5001: B7/B38/B40/B41 PA
5. U4912: SPDT SW. for B38(B41)/B7
6. U4909: LNA for B38/B41/B7
7. U4501: Transceiver(WTR2965_1)
8. U5118: B7 DRx SAW
9. U5119: SPDT SW. for B7/B41(B38)
10. U5111: LNA for B7/B38/B41
11. U5110: B3/B7 Diplexer
12. U4704: Div Ant. SW.



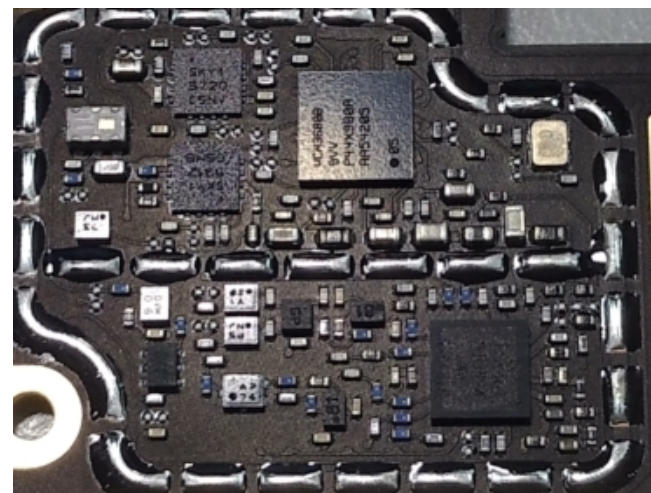
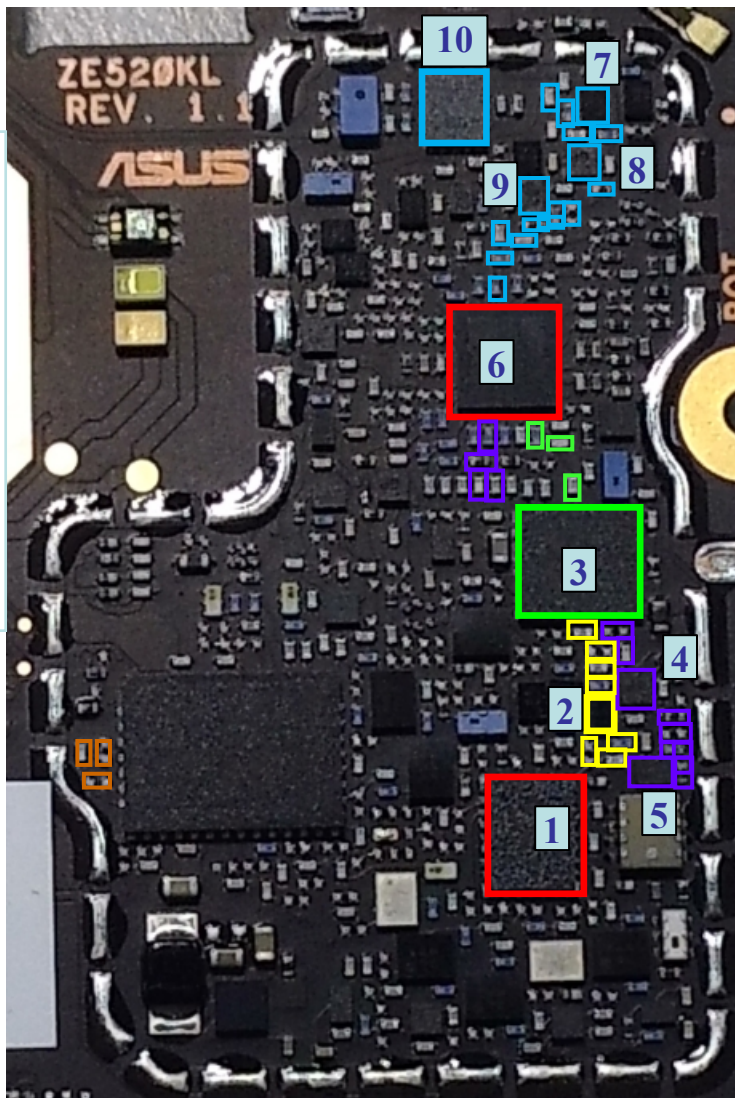
RF (Top)

- TRx common comp.
- B7 Tx Path
- B7 TRx Path
- B7 PRx Path
- B7 DRx Path

B38/B41 TRx/DRx

RF (Bottom)

1. U4701: Primary Ant. SW.
2. U5005: B38/B41 TRx SAW
3. U5001: B7/B38/B40/B41 PA
4. U4912: SPDT SW. for B38(B41)/B7
5. U4909: LNA for B38/B41/B7
6. U4501: Transceiver(WTR2965_1)
7. U5109: B38/B41 DRx SAW
8. U5119: SPDT SW. for B7/B41(B38)
9. U5111: LNA for B7/B38/B41
10. U4704: Div Ant. SW.



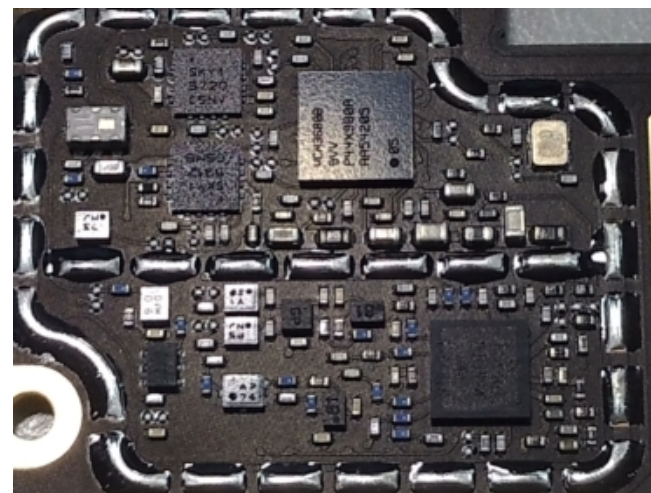
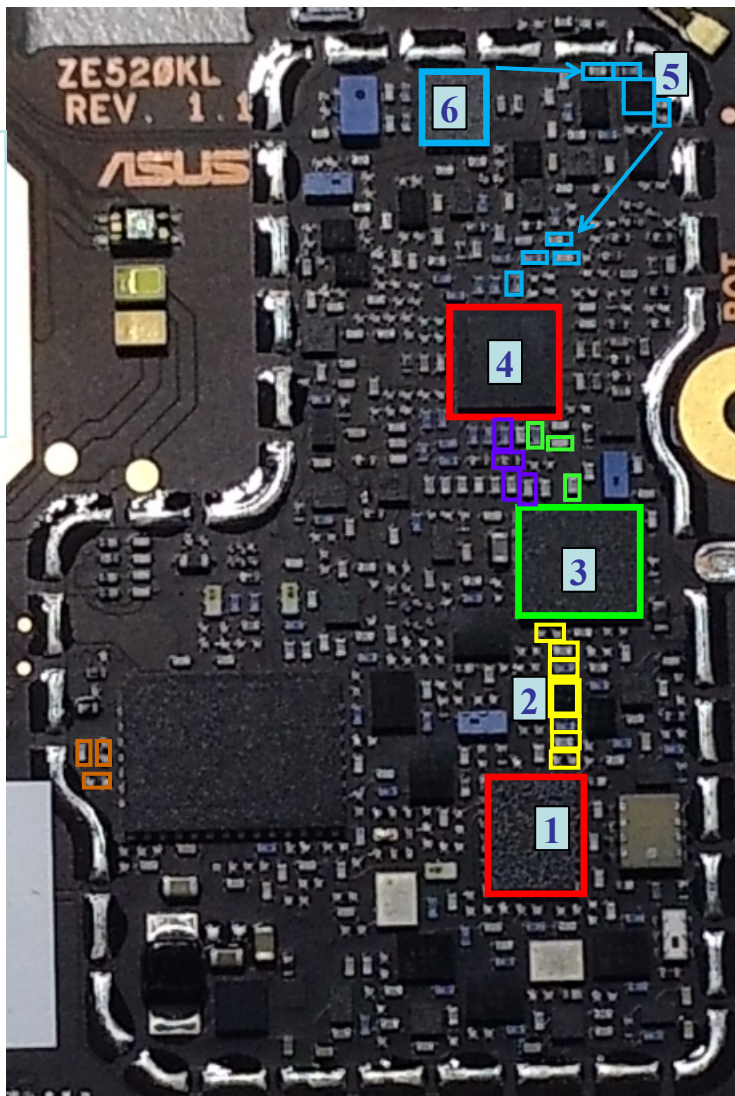
RF (Top)

- TRx common comp.
- B38/B41Tx Path
- B38/B41 TRx Path
- B38/B41 PRx Path
- B38/B41 DRx Path

B40 TRx/ B40 DRx

RF (Bottom)

1. U4701: Primary Ant. SW.
2. U5002: B40 TRx SAW
3. U5001: B7/B38/B40/B41 PA
4. U4501: Transceiver(WTR2965_1)
5. U5108: B40 DRx SAW
6. U4704: Div Ant. SW.



RF (Top)

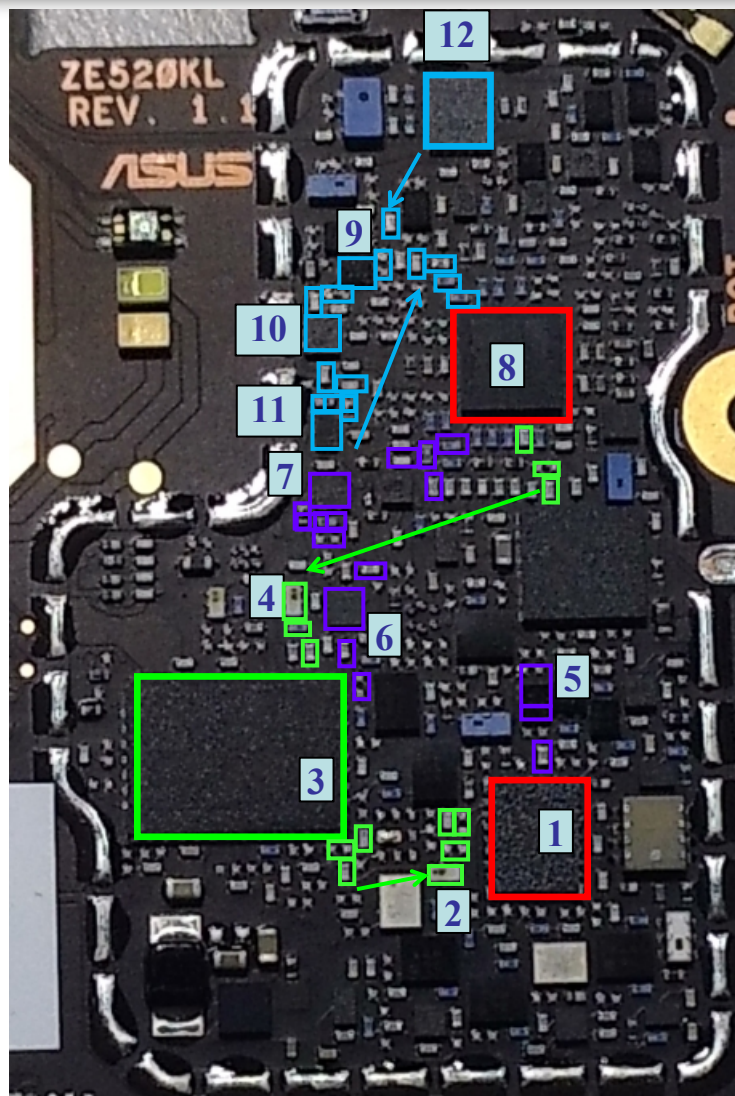
- TRx common comp.
- B40Tx Path
- B40 TRx Path
- B40 PRx Path
- B40 DRx Path

B39 TRx/DRx

RF (Bottom)

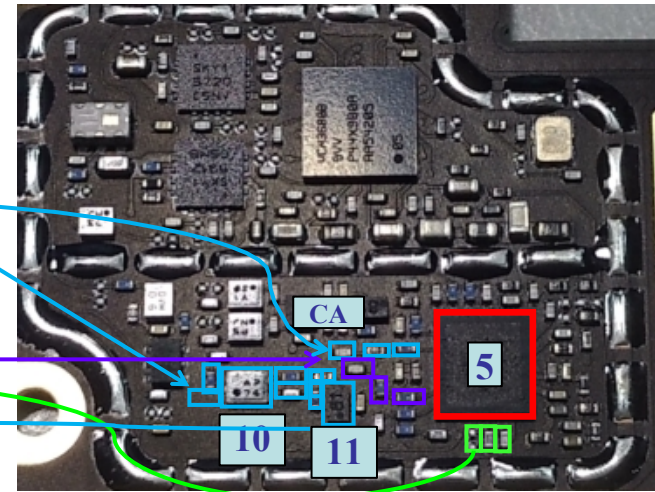
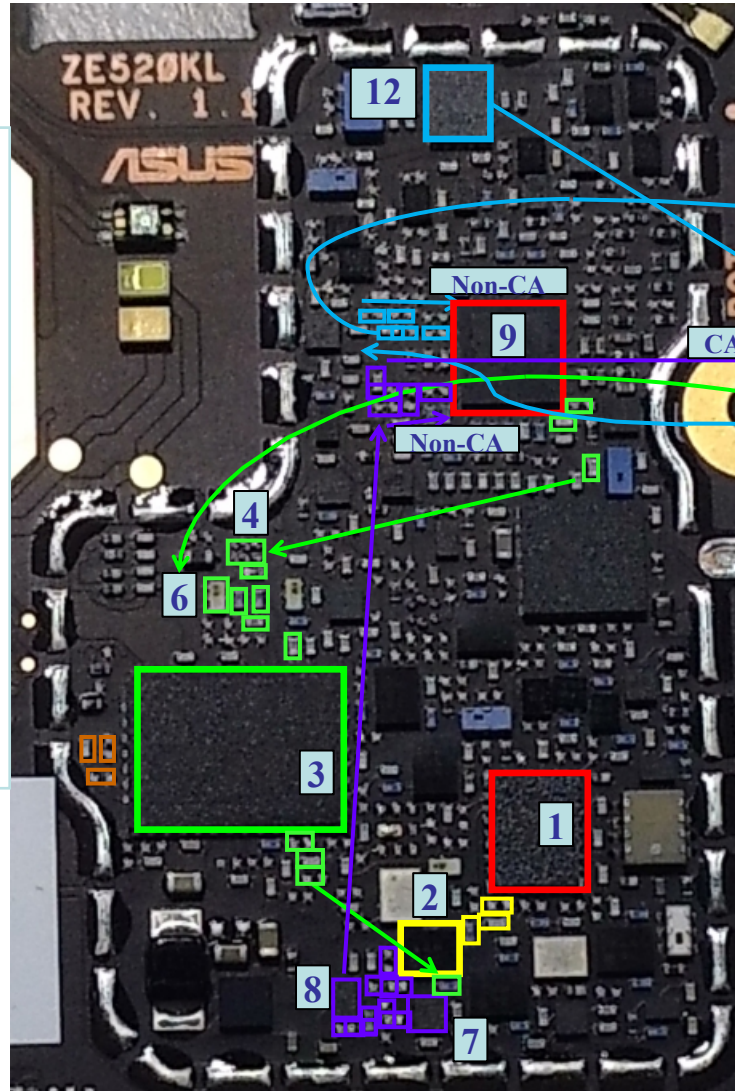
1. U4701: Primary Ant. SW.
2. U4801: B39 Tx SAW
3. U4805: MMPA(MB/LB)
4. U4803: LPF for B1/B2/B3/B4/GHB/B39
5. U4905: B39 PRx SAW
6. U4904: SP3T SW. for PRx B2/B3/B39
7. U4908: LNA for B2/B3/B39
8. U4501: Transceiver(WTR2965_1)
9. U5107: B39 DRx SAW
10. U5103: SP3T SW. for DRx B2/B3/B39
11. U5116: LNA for B2/B3/B39
12. U4704: Div Ant. SW.

-  TRx common comp.
-  B39 Tx Path
-  B39 PRx Path
-  B39 DRx Path



B28A TRx/ DRx

RF (Bottom)

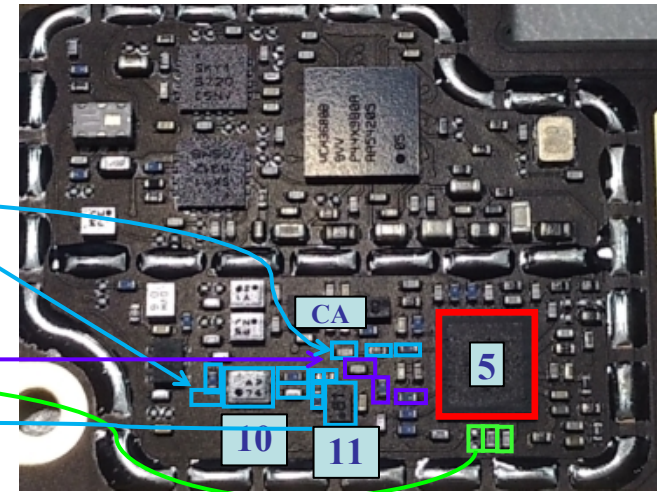
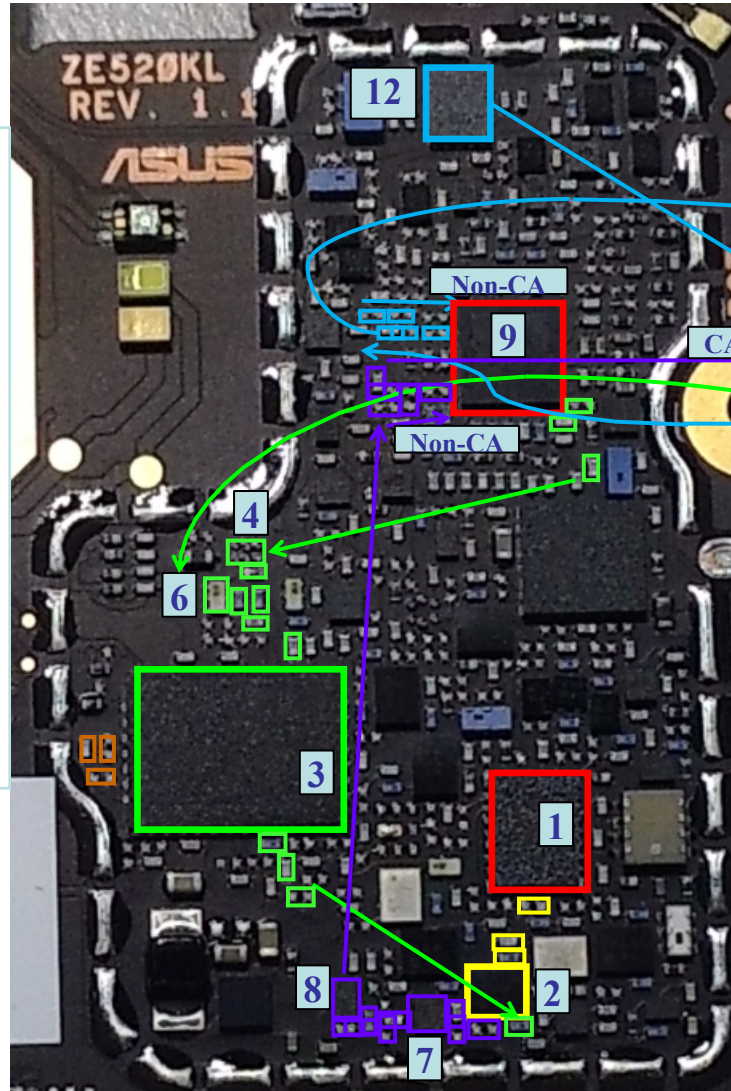


RF (Top)

- TRx common comp.
- B28A Tx Path
- B28A TRx Path
- B28A PRx Path
- B28 DRx Path

B28B/B20 TRx/ DRx

RF (Bottom)

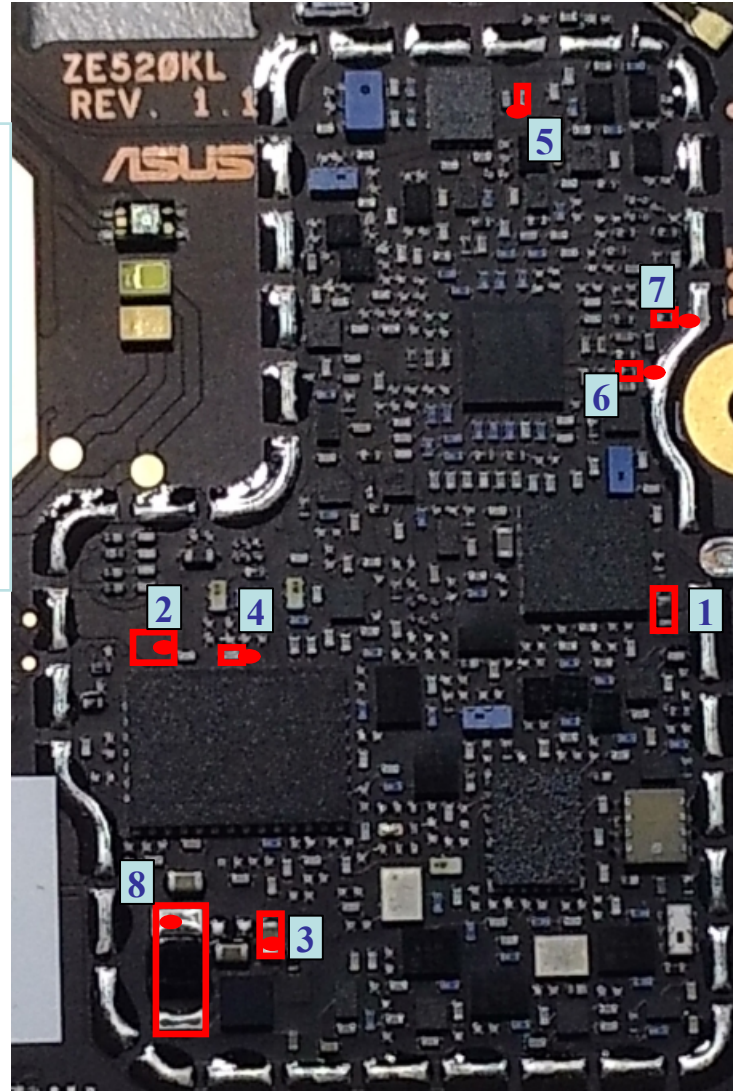


RF (Top)

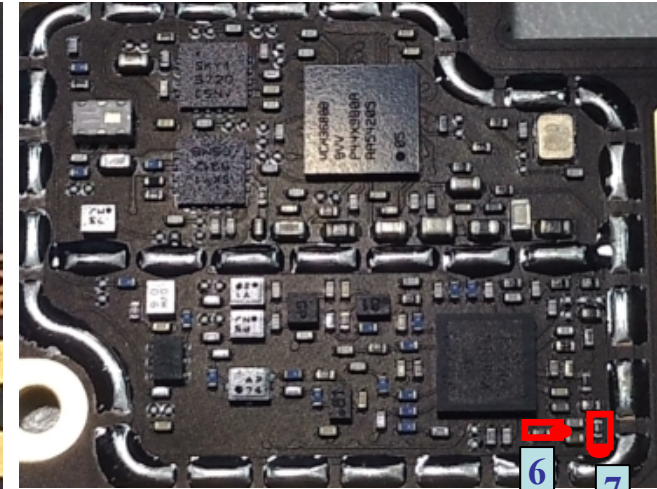
- TRx common comp.
- B28B/B20 Tx Path
- B28B/B20 TRx Path
- B28B/B20 PRx Path
- B28/B20 DRx Path

Voltage Measurement

RF (Bottom)

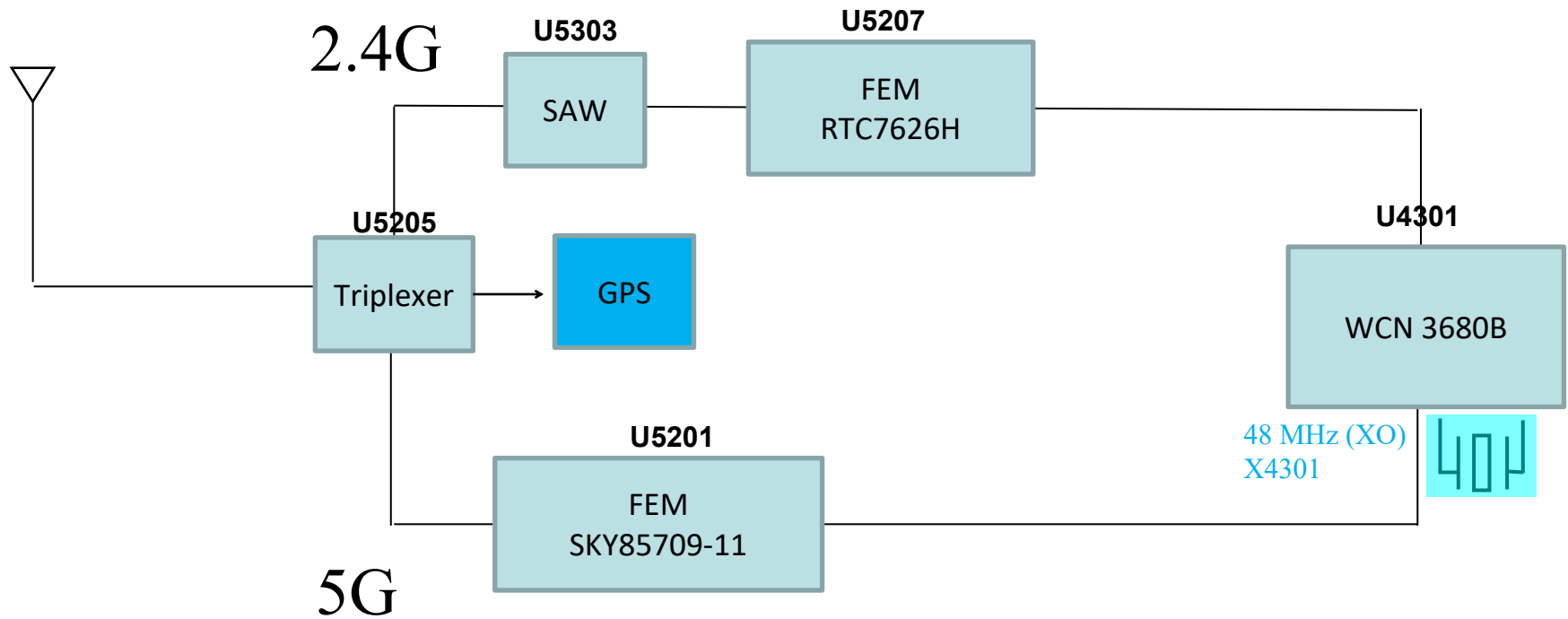


1. R4289.2 High Band PA → 3.8V
2. C4812.1 2G PA → 3.8V
3. C4601.1 DCDC power in → 3.8V
4. C4802.1 MIPI 1.8V → 1.8V
5. C4709.1 VDD_SWITCH → 2.85V
6. R4404.2(R4502.2) Transceiver
VDD RF2 → 1.8V
7. R4405.2(R4505.2) Transceiver
VDD RF1 → 1.05V
8. L4601.2 DCDC output

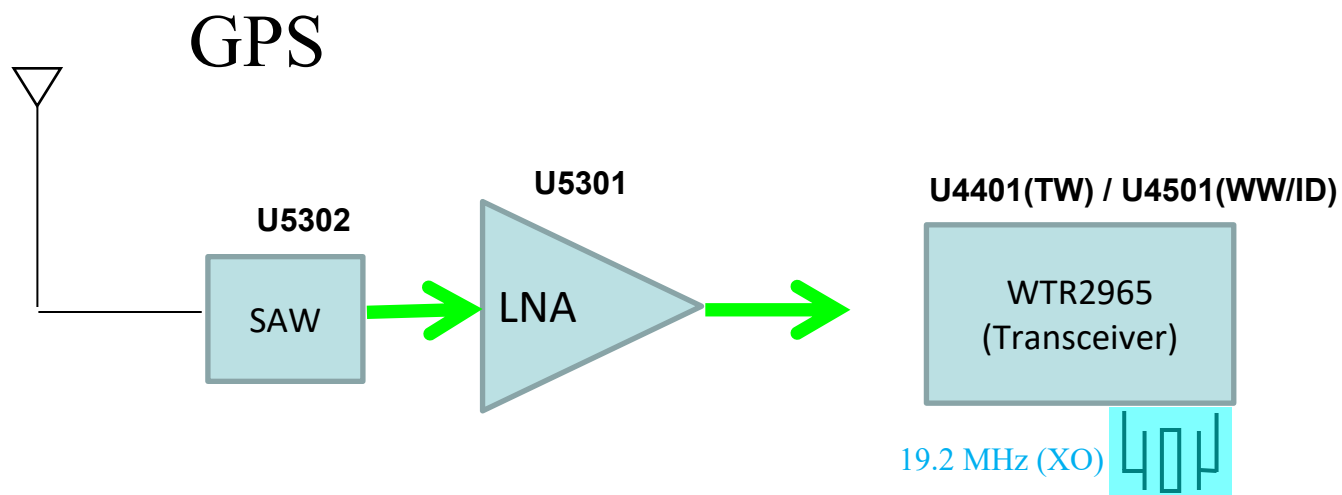


RF (Top)

Wifi/BT Block Diagram

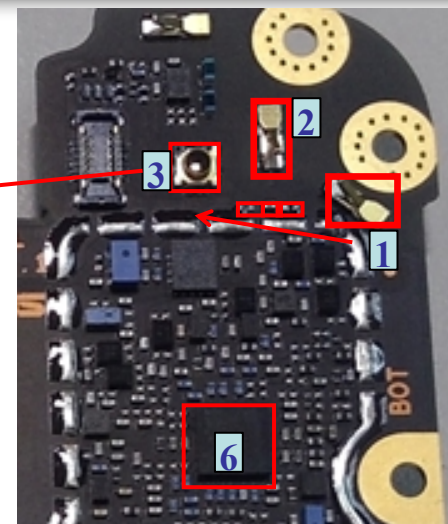
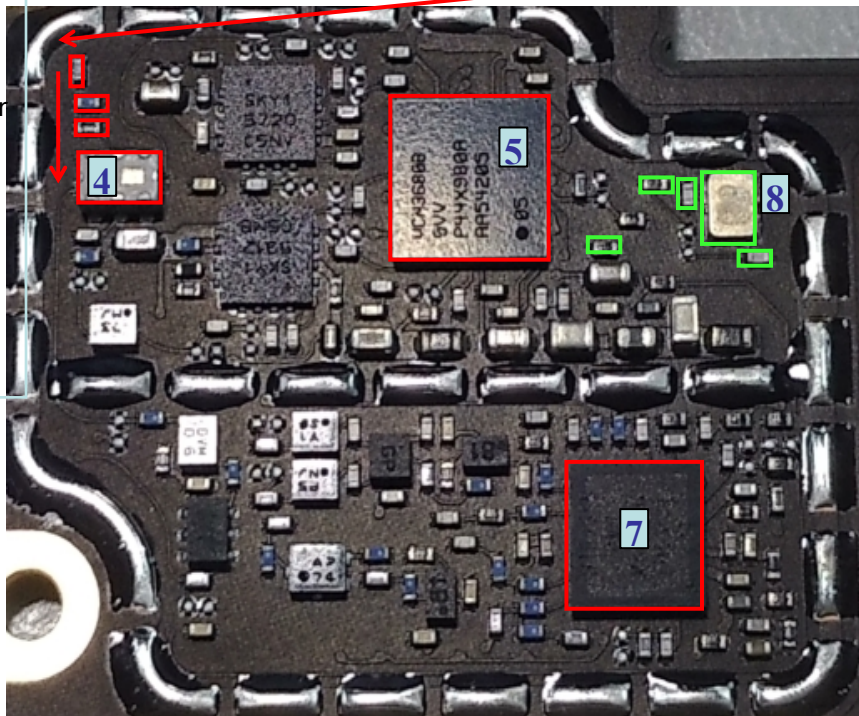


GPS Block Diagram



Wifi/BT/GPS Common

RF (Top)



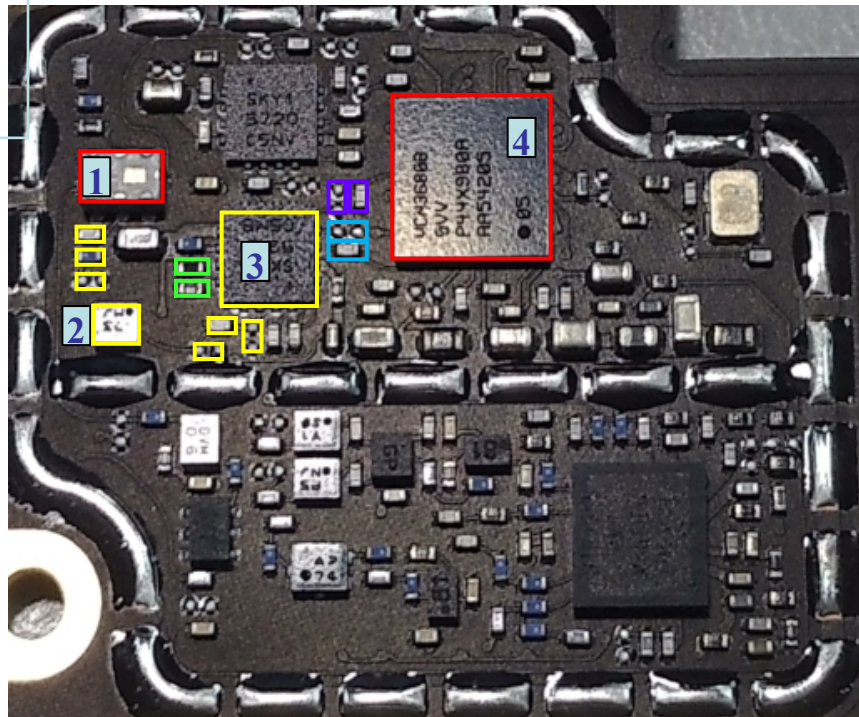
RF (Bottom)

- TRx common comp.
- 48MHz Xtal

Wifi 2.4G/BT

RF (Top)

1. U5205: WIFI2.4G/5G/GPS Triplexer
2. U5203: WIFI2.4G(BT) SAW
3. U5207: WIFI2.4G/BT FEM
4. U4301: WIFI Chip(WCN3680B)

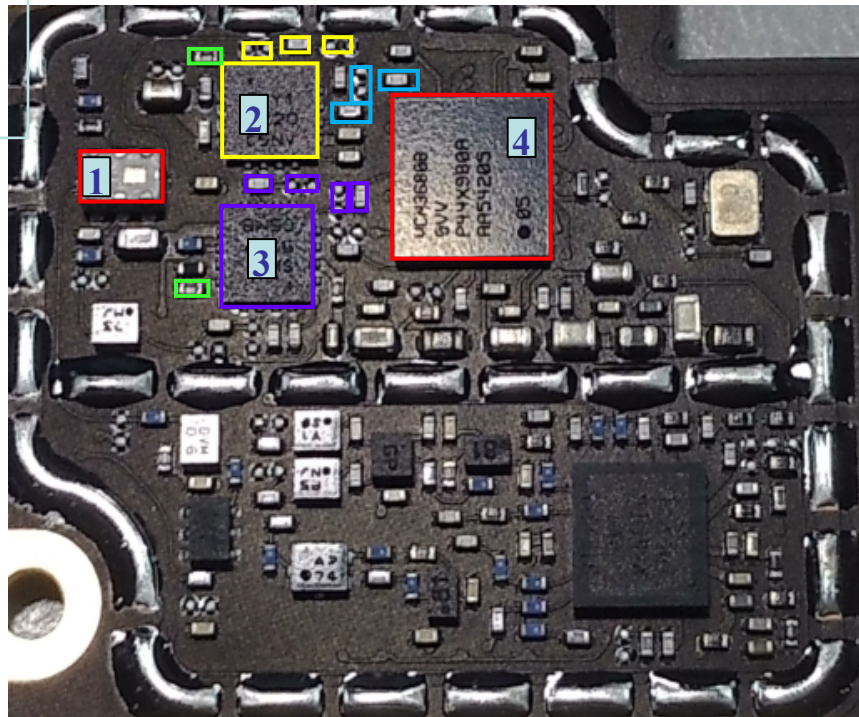


- TRx common comp.
- 2.4G PDET Path
- 2.4G+BT TRx Path
- 2.4G PA in
- BT PA in

Wifi 5G Tx/Rx

RF (Top)

1. U5205: WIFI2.4G/5G/GPS Triplexer
2. U5201: WIFI 5G FEM
3. U5207: WIFI 2.4G FEM
4. U4301: WIFI Chip(WCN3680B)

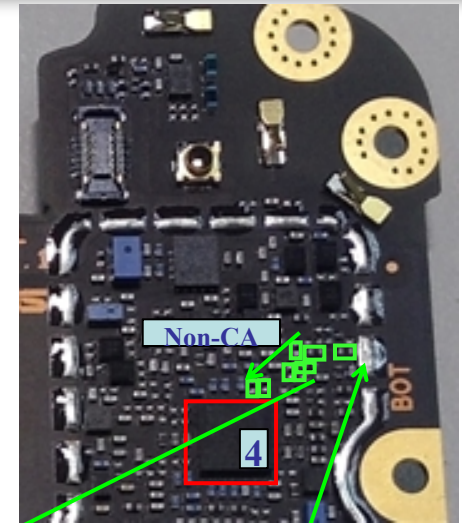
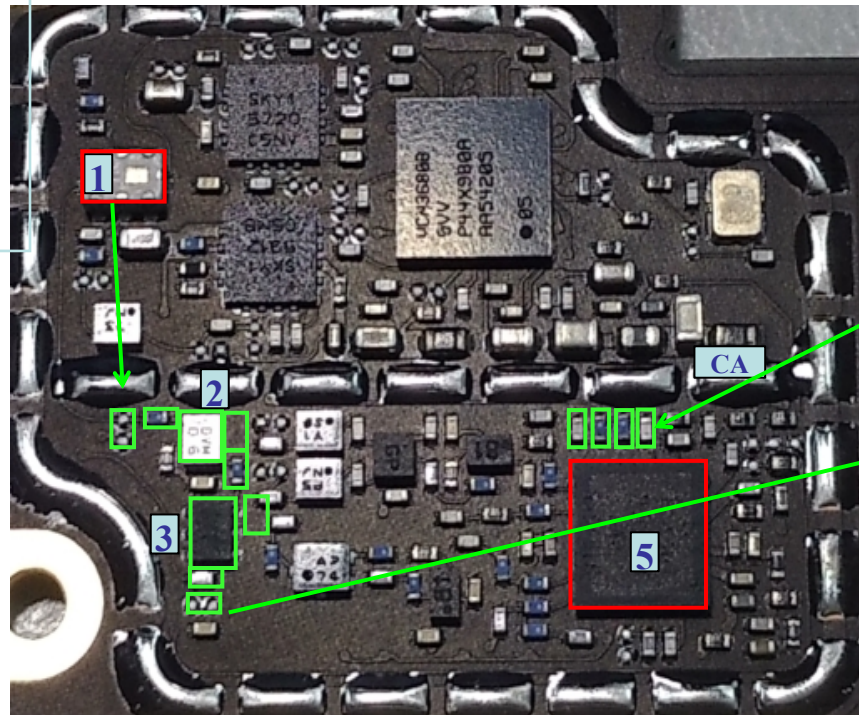


- TRx common comp.
- 5G PDET Path
- 5G TRx Path
- 5G Tx PA in
- 5G Rx Path

GPS

4 RF (Top)

1. U5205: WIFI2.4G/5G/GPS Triplexer
2. U5302: GPS PreSAW
3. U5301: GPS LNA
4. U4501: Tranceiver(WTR2965_1)
(GPS for WW/ID SKU)
5. U4401: Tranceiver(WTR2965_0)
(GPS for TW SKU)



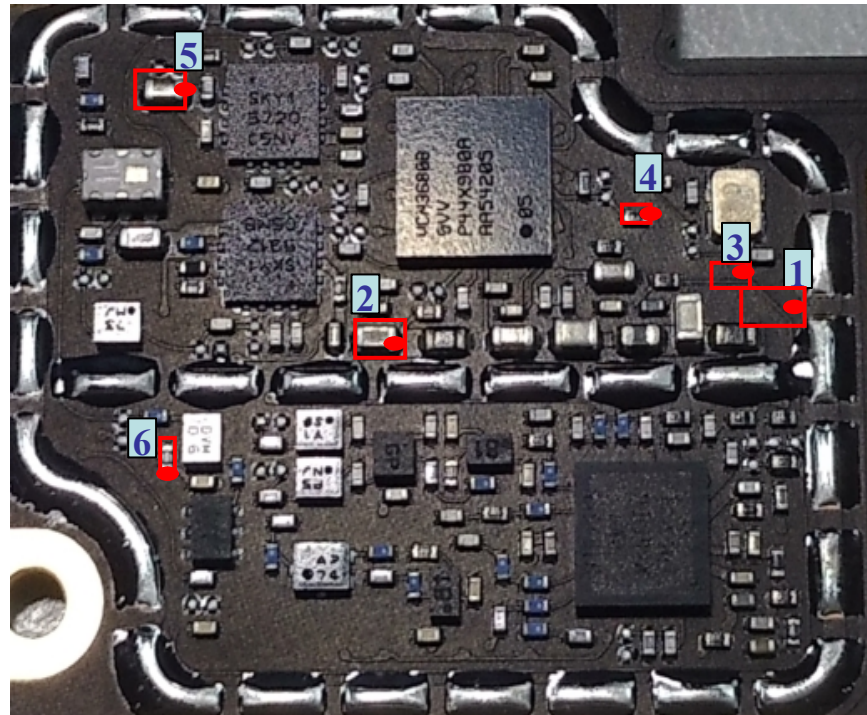
RF (Bottom)

-  TRx common comp.
 GPS Path

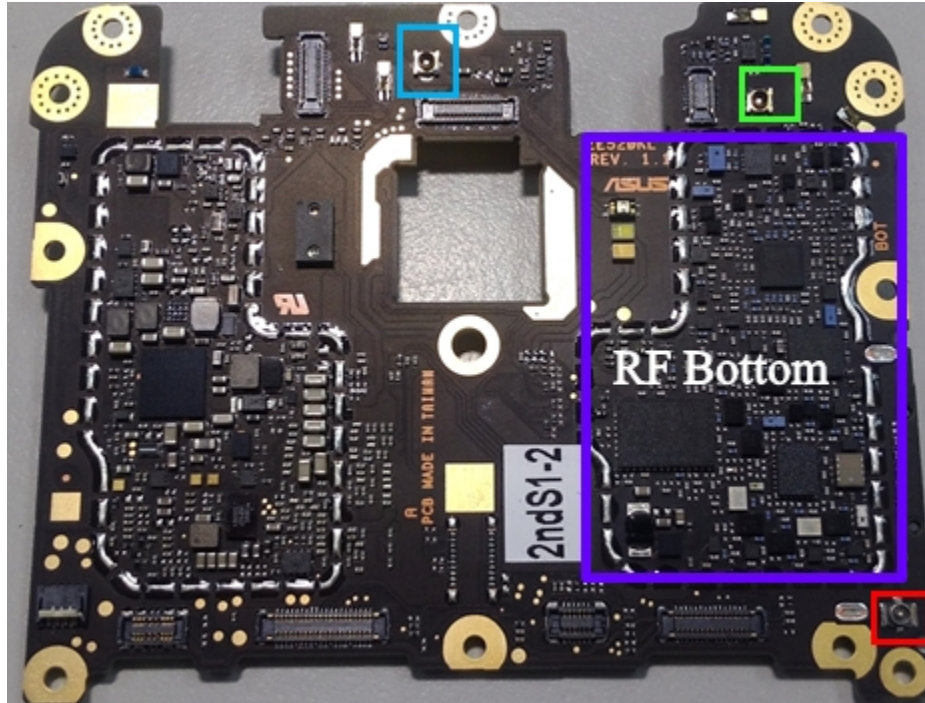
Voltage Measurement

RF (Top)

1. R4302.1 VREG_L19_1P35 →1.3V
2. C4302.1 VDD_WCN_3P0 →3.3V
3. R4306.1 VREG_L7_1P8 →1.8V
4. R4307.1 VREG_L5_1P8 →1.8V
5. C5206.1 VPH_PWR_WL(FEM power) →3.8V
6. L5301.1 VDD_GPS_LNA →1.8V



ZE520KL RF Cable Spec



-  Main TRx RF Connector
-  Diversity Rx RF Connector
-  WIFI/BT/GPS Connector



MURATA_11421
5000009816.pdf

Phone-Diversity cable --> **Murata:MXHQ87WJ3000**

Phone-Main cable --> **Ipex: MHF4 cable**

WiFi cable --> **Murata:MXHQ87WJ3000**



MHF4 VSWR &
insertion Loss Rev. 1.pdf

Thanks for your attention!